

CORE CURRICULUM



Operations Management

Roy D. Shapiro, Series Editor

READING

Operations Strategy

.....
MARCO IANSITI

HARVARD BUSINESS SCHOOL

ALAIN SERELS
.....

8000 | Revised: June 1, 2024

Table of Contents

1 Introduction	3
2 Essential Reading	6
2.1 A Brief History of Operations Strategy	6
2.1.1 Global Competition and the New Focus on Operations.....	7
2.1.2 What Is Operations Strategy?	7
2.2 Strategy and Operations.....	8
2.2.1 Directed Strategy.....	8
2.2.2 Emergent Strategy	10
2.2.3 Strategic Impact of Operations	11
2.3 Focus	11
2.3.1 Generic Operations Performance Objectives	12
2.3.2 Focus Is Restricting the Range of Activities	13
2.3.3 The Benefit of Focus.....	13
2.3.4 Loss of Focus.....	15
2.4 The Operations Strategy Framework	15
2.4.1 Resource-Based Operations Strategy Framework.....	16
2.4.2 The Operations Strategy Matrix.....	18
2.4.3 Assessing the Operations Strategy	20
2.4.4 Operations as a Complex System.....	22
2.5 Operations Decision Areas.....	22
3 Supplemental Reading.....	25
3.1 Managing Disruptive Strategy Transitions: Microsoft Server and Tools.....	25
3.1.1 STB's Traditional Operations Strategy	25
3.1.2 A Disruptive Change in STB Strategy	27
3.1.3 Results	29
4 Key Terms	31
5 For Further Reading.....	32
6 Endnotes	33
7 Index	33

Marco Iansiti, David Sarnoff Professor of Business Administration, Harvard Business School, coauthored this core reading with the assistance of Alain Serels, HBS Research Associate, 2011–2013.

Copyright © 2013, 2024 Harvard Business School Publishing. All rights reserved. To order copies or request permission to reproduce materials (including posting on academic websites), call 1-800-545-7685 or go to <http://www.hbsp.harvard.edu>.

1 INTRODUCTION

A company's operations are crucial to its strategic differentiation and performance. In this reading, you will discover why this is so and how to make sure that an operation's components are aligned with the company's strategic direction. The key to such alignment is developing a company's **operations strategy**. Let's first take a step back and understand what modern operations entail and then break down the term *operations strategy* into its component parts.

A dictionary definition of operation is "power to act."¹ This definition provides motivation, even inspiration, for a broad and general view of operations as the area that drives action in the firm. The definition is particularly helpful because it involves not only action (what businesspeople refer to as business execution) but also the power to make effective action a reality.

The operations function organizes the systems of resources that create and deliver the firm's product or service offerings. A firm's operations are centered on its resources engaged in product development, manufacturing, service management, supply chain management, channel management, and other execution-oriented functions that act to create and deliver value to customers. These operational resources can transform physical raw materials (like aluminum that is transformed into soda cans), information resources (like trends data that are transformed into forecasts), financial resources (as in an investment that is transformed into a production process), or customers (like a patient with a cavity who is transformed into a patient with a dental filling). The collection of such transformative resources (physical, information-based, financial, and human), together with the system by which they are organized, make up the operations function.

Strategy, on the other hand, refers to a framework or plan for the utilization of resources. A strategy can be top-down directive, defined by senior leaders to guide execution. It can also be bottom-up emergent, inferred implicitly from the pattern of decisions and actions taken by people throughout an organization. See **Exhibit 1**.

When we combine operations and strategy to construct the term *operations strategy*, we arrive at a "framework or plan for the utilization of resources that drive business execution." At its core, operations strategy serves as the crucial link between the planned strategy and the executed reality.

EXHIBIT 1 Operations Strategy Connects Directed and Emergent Strategies

Traditionally, the field of operations focused on the manufacturing sector—companies like Ford, Boeing, and General Electric that produced tangible products in company-owned factories. In fact, topics in this area were often defined by terms such as factory management, production management, and manufacturing strategy, emphasizing a factory-centered view of the world. During the 1980s, the field was expanded to include service organizations. Private-sector service firms have been the largest sector of the US economy since the end of World War II, with services now representing more than 70 percent of total gross domestic product (GDP) in the United States.² In China, long referred to as “the world’s factory,” services accounted for just 41 percent of the country’s GDP in 2005. Six years later, more people were employed in services than in manufacturing (35 percent versus 29 percent), and from 2015 to 2022, services accounted for over half of GDP.³

As we shall see, despite some key differences, many of the issues faced by manufacturers are similar to those faced by service providers such as an airline, a retailer, or a restaurant chain. We can still speak about operational complexity, focus, impact of information, and human-based resources, and so on.

Going well beyond traditional services, the past quarter century has witnessed a dramatic increase in firms with novel, knowledge-based business models, which require alternative forms of operations. For example, low-capital service firms, such as consulting firms and investment banks, have few physical assets but rely on employees and accumulated knowledge to deliver value to customers. Virtual corporations—such as Li & Fung, which has orchestrated global supply chains for retailers such as Walmart and Target, as well as brands such as Nike and Tommy Hilfiger—outsource nearly all operating functions. These firms are becoming increasingly essential to the economy. Organizations based on intellectual property (IP), including application developers or Software as a Service (SaaS) providers like Microsoft; platform-based service providers such as Airbnb; and patent portfolio managers like Intellectual Ventures, produce knowledge or virtual

goods at little marginal cost and employ a large portion of their resources in the generation or acquisition of new IP.

Even though they are drastically different from conventional manufacturers and service providers, the new generation of knowledge-based firms also experiences operating challenges and therefore benefits from thinking through operations strategy. Their decisions about physical, human, information-based, and financial resources open (or close) strategic options just like decisions in manufacturing firms.

The key enabler in the spread of knowledge-based business models is the development of communications and information technologies. The so-called information economy has enabled significant changes in the way companies operate and has defined new classes of resources. Four trends in particular highlight the importance of operations strategy in the new economy:

1. *Focus on knowledge-based assets.* IP, virtual property, know-how, and trade secrets have always existed, but such knowledge-based assets have become increasingly critical to competitiveness and differentiation for many of the world's most innovative companies. Information technology (IT) companies like Microsoft and Apple develop proprietary technologies that form the foundations for state-of-the-art products and services. Biotech firms such as Genentech and Genzyme develop new compounds and patented production techniques to bring new medicines to market.
2. *Data as an asset.* The information collected from trillions of searches has been one of Google's most strategic assets, giving the company a competitive advantage through improved search results and the increased effectiveness of targeted advertisements. Competitors could develop their own search algorithms, tools, and infrastructure, but replicating Google's data set has been a greater challenge. The emergence of generative artificial intelligence (AI) tools and general-purpose chatbots is not only a disruptive search technology; AI has also opened up new possibilities for companies to use their own, often previously untapped, data sets. For example, Australia's Commonwealth Bank used AI to create "self-learning 'Adaptive Decision Models (ADMs)'" designed to assess, in real time, "which messages resonated most with which types of customers," and tailor services, interactions, products, and services to their needs.⁴ In industries as diverse as manufacturing, electric utilities, and health care, AI offers companies opportunities to better access and understand their data and to use it in new ways.
3. *Globalization.* In an increasingly global economy, many companies supply markets in every country where sufficient demand exists, cooperating with partners wherever they may be found and searching out efficient labor and talent in all parts of the world. The spread of communication technologies continues to make the world even smaller as firms can increasingly identify demand and coordinate activities around the globe.
4. *Distributed operations and outsourcing.* Where once firms developed large industrial parks and concentrated all operations in a single location, distributed operations have become the new norm. The first level of distribution occurred geographically, with the regional distribution of activities and the establishment of secondary manufacturing sites distant from initial facilities. The second level occurred with the modularization of components, which allowed for the production of separate components in separate sites or even by separate suppliers. More recently, a third level of distribution has

modularized the process of operations and the outsourcing of manufacturing across multiple suppliers.

Consider, for example, the steps involved in the manufacture of a mobile phone: the phone may be designed by Samsung in South Korea; the microprocessor developed by ARM in the United Kingdom and produced by Mediatek in Taiwan; the chipsets developed by Qualcomm in the United States; the operating system developed by Google in the United States; the components assembled by factories in Vietnam, India, or Brazil; and the apps produced by thousands of independent software vendors around the world. The development and production of one product can now routinely include thousands of companies in dispersed locations, each contributing a critical element.

In sum, the operations function of a modern firm can be very different from the loud, crowded factory we all may have once pictured. Although companies take many forms, the principles and frameworks developed here should be broadly applicable. Every company has operations; therefore, every company should better understand its operations strategy.

This reading provides tools for understanding, developing, and analyzing a company's operations strategy, beginning with an introduction to the concept and study of operations strategy. Section 2.2 introduces the reader to the concept of corporate strategy and its connection with the long-term objectives of the operations function. Section 2.3 addresses the issue of focus in operations and examines generic operations performance objectives. Section 2.4 develops an operations strategy framework as a tool to ensure alignment between actions and objectives. Section 2.5 presents a list of management decision areas. The Supplemental Reading in section 3 provides an example of applying the operations strategy framework through an in-depth case study.

2 ESSENTIAL READING

2.1 A Brief History of Operations Strategy

Operations strategy has an important tradition. The field was built to fight one of the great paradoxes of modern management. In many firms, especially traditional manufacturing firms, most resources are used to drive operating tasks. Operating decisions frame options for strategy and define the capabilities that an organization has at its disposal. Few decisions are as critical. Until the late 1960s, however, operations were consistently thought of in a tactical and reactive fashion, with little perceived strategic value.

In 1969, Harvard Business School professor Wickham Skinner introduced the concept of an operations strategy in an article titled “Manufacturing—Missing Link in Corporate Strategy.”⁵ He observed that senior managers took a hands-off approach to the operations function, often delegating decision making to technical specialists. Management, he explained, looked at operations as a black box and did not consider the strategic importance of operations decisions:

Frequently the interrelationship between production operations and corporate strategy is not easily grasped. The notion is simple enough—namely, that a company’s competitive strategy at a given time places particular demands on its manufacturing function, and, conversely, that the company’s manufacturing posture and operations should be specifically designed to fulfill the task demanded by strategic plans.⁶

Operations, Skinner argued, was central to achieving the firm’s goals and therefore was of crucial strategic importance:

The purpose of manufacturing is to serve the company—to meet its needs for survival, profit, and growth. Manufacturing is part of the strategic concept that relates a company’s strengths and resources to opportunities in the market. Each strategy creates a unique manufacturing task. Manufacturing management’s ability to meet that task is the key measure of its success.⁷

Skinner encouraged managers to think of operations as the implementer of strategy. For a strategy to succeed, the key operations decisions should be thought of as strategic and should therefore draw the attention of senior management.

2.1.1 Global Competition and the New Focus on Operations

After World War II, US-based firms faced little competition in their domestic market. The war had brought a push to generate operating efficiencies by implementing mass production techniques in the manufacturing of wartime goods. The most capable competitors, based in Europe and Japan, had been hampered by the war and posed little threat. But by the middle of the 1960s, imports to the United States began to increase rapidly as foreign competitors rebuilt their production bases. By 1971, imports exceeded exports, and the United States has run a net deficit in exports every year since 1976. As overseas competitors consistently gained ground on local firms in the US market, senior managers began to focus on the activities of the operations function.

2.1.2 What Is Operations Strategy?

Operations strategy is a framework or plan for the utilization of resources that drive business execution. It gives managers a way to see how operations activities fit with the

company's overarching goals and objectives. The aim is to ensure strategic alignment between operations' resources and the strategic objectives of the firm.

In practice, creating a new operations strategy or evaluating an existing one occurs in two distinct steps:

1. Define the long-term objectives of operations. The first step, sometimes called the process of operations strategy, is to define and prioritize a specific mission with a set of concrete objectives for the operations function.
2. Develop a plan to meet the objectives. Once the objectives are clear, the second step, occasionally referred to as the content of operations strategy, is to create a plan for fulfilling the objectives. This plan guides management by describing the pattern of decisions and actions that develop, organize, and utilize resources. This process can be run in reverse, to check that the choices that have been made will allow the operations function to achieve its objectives.

Consider the Swiss pharmaceutical company Novartis. You might assume that its main operations activity is the manufacture and distribution of pharmaceuticals. For Novartis and other large pharmaceutical companies, however, the main activity is drug discovery and development. In the early 2000s, Novartis was finding it increasingly difficult and expensive to discover and develop new drugs. The pharmaceutical development team first defined the objective to expand capacity and improve research and development (R&D) efficiency. Next, the team developed a plan to centralize its research activities near top universities and hospitals and to partner with outside laboratories. As the company implemented his plan, Novartis's pharmaceutical pipeline increased from 75 compounds in 2004, to 138 in 2012, to more than 230 by 2020.⁸

2.2 Strategy and Operations

In its most basic sense, an operations strategy defines the long-term direction of the firm's operating resources. We can also think of marketing strategy, finance strategy, corporate strategy, or the strategy of any other organizational unit. These strategies are not independent because the choices of each unit can affect the entire company. In this section, we explore the link between the strategy of the operations function and that of the company as a whole.

2.2.1 Directed Strategy

One definition of strategy is based on Michael Porter's work: it is "an integrated set of choices that positions a firm in an industry so as to generate superior financial returns over the long run."⁹ In this sense, strategy can be thought of as the plan for managing internal resources that will allow a company, given the external environment in which it operates, to best meet stakeholders' objectives. This view of strategy is often called **directed**

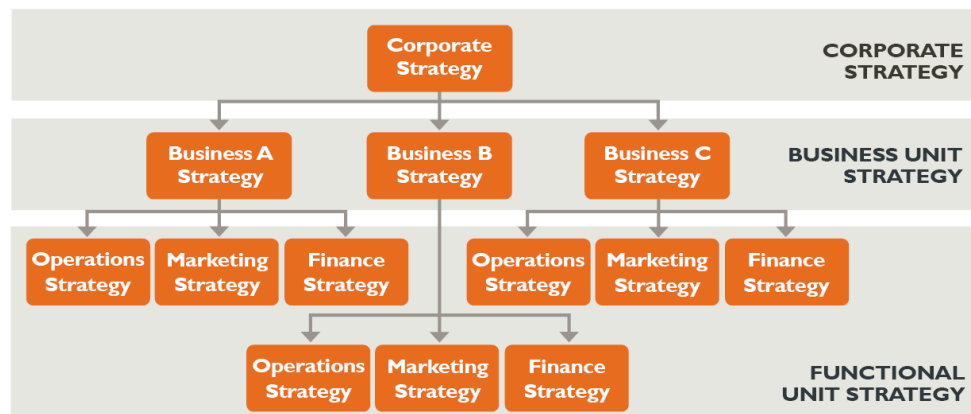
strategy or **deliberate strategy** because it is a plan developed at the top of the organization by senior management and disseminated to all those involved in implementation and execution.

In this top-down directed model, strategy follows a clear path through the organization, called the strategy hierarchy (see **Exhibit 2**). At the top, **corporate strategy** is concerned with the most aggregate, high-level decisions concerning the selection of industries, the configuration of business units, the allocation of resources, and the expected returns and growth from the businesses.

The next level in the strategy hierarchy is **business unit strategy**, which each business unit must develop to determine how it will compete in its market given the corporate strategy. Business unit strategy decisions include the choice of position to occupy in the industry and competitive advantages to develop. When a company has only one business unit, corporate strategy and business unit strategy are one and the same.

Each function within a business unit creates a **functional strategy** to implement the business unit strategy. Operations, marketing, finance, and other functions translate the desired market positioning and competitive advantage into function-specific competitive priorities and performance objectives, and develop a plan for achieving them. In this view, operations strategy is the translation of the business unit strategy into an actionable plan for the development, organization, and utilization of operations' resources to best meet the performance objectives derived from the business unit strategy.

EXHIBIT 2 Strategy Hierarchy



Source: Reprinted and adapted by permission from Robert H. Hayes and Steven C. Wheelwright, *Restoring Our Competitive Edge: Competing Through Manufacturing* (New York: John Wiley & Sons, 1984), p. 28. Copyright © 1984 by John Wiley & Sons.

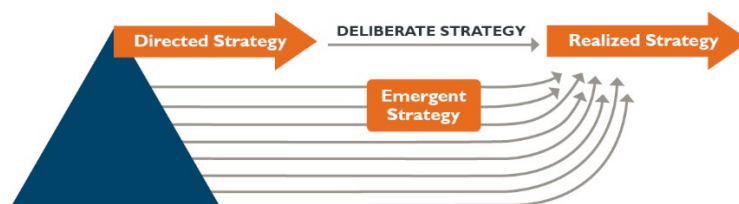
We can clearly see these three levels of strategy at work in the Walt Disney Company. Recognized by children all over the world as the creator of Mickey Mouse and other classic children's entertainment, Disney is a complex corporation with many activities. Disney's

corporate strategy has determined the segments of the media and entertainment market where the company can achieve competitive advantage. During that period, Disney added attractions to existing theme parks, constructed a Disney-themed resort in Shanghai, and added to the fleet of cruise ships. Its operations strategy supported the business strategy by focusing on customer satisfaction at existing attractions while supporting new opportunities for expansion. By 2023, after this period of growth and expansion globally in Europe and Asia, Disney's business units were restructured from five to three core, global business units—entertainment, sports, and experiences—to “return creativity to the center of the company, increase accountability, improve results, and ensure the quality of our content and experiences.”¹⁰ Disney Experiences, which includes theme parks, hotels, resorts, cruise ships, related consumer merchandise, and licensing revenues, continues to generate almost half of Disney's \$88 billion in revenue.

2.2.2 Emergent Strategy

Operational execution can occur in any function or business unit. Managers throughout the organization frequently make decisions relating to the development, organization, and utilization of resources. The resulting pattern of decisions constitutes the **emergent strategy** or **realized strategy** of the firm because those decisions determine market position, competitive advantage, and performance (see **Exhibit 3**). For example, to generate additional sales, US close-out retailer Dollar General began offering household consumable products in the 1990s. Customer response was so strong that the company gradually expanded the new offering. By 2020, consumables made up 79 percent of Dollar General's \$33 billion in sales across more than 16,000 stores—even though top management had not planned this shift in strategy from the outset.¹¹

EXHIBIT 3 Directed and Emergent Strategies



Source: Adapted with permission from Henry Mintzberg, "Patterns in Strategy Formation," *Management Science* 24, no. 9 (1978): 945. Copyright © 1978, the Institute for Operations Research and the Management Sciences, 7240 Parkway Drive, Suite 300, Hanover, Maryland 21076.

Emergent strategies define the firm's actual strategy. Even when a directed strategy is developed, the pattern of implementation decisions may not be in line with it. In some cases, lower-level managers may misinterpret the directed strategy or intentionally diverge from it. In other cases, the directed strategy may not be communicated effectively to them. Even when managers understand the directed strategy, they may be faced with decisions that have little apparent strategic impact but collectively lead to unintended market positioning and performance.

Normal pressures on operating managers can lead to decisions that don't fit with the objectives of the directed strategy. Marketing, for instance, in trying to better serve customers, will naturally pressure operations to provide more flexibility or quicker turnaround times. Finance will urge operations to reduce costs and increase efficiencies. Pressures also develop within operations: Quality control managers may push for increased quality, process engineers for increased automation, and industrial engineers for increased standardization. The realized operations strategy can be heavily influenced by the pattern of responses to these pressures.

2.2.3 Strategic Impact of Operations

In essence, the firm's collection of resources determines its ability to attain a specific competitive position. Therefore, operations decisions that affect the systems of resources have a direct effect on firm performance and can influence the future strategy of the company as a whole. As Wickham Skinner noted in his 1969 article:

The connection between manufacturing and corporate success is rarely seen as more than the achievement of high efficiency and low costs. In fact, the connection is much more critical and much more sensitive. Few top managers are aware that what appear to be routine manufacturing decisions frequently come to limit the corporation's strategic options, binding it with facilities, equipment, personnel, and basic controls and policies to a noncompetitive posture which may take years to turn around.¹²

Skinner realized that operating decisions can constrain higher-level strategic options. Robert Hayes and Steven Wheelwright took this argument one step further, arguing that operations' systems of resources can be seen as a strength on which competitive advantage can be built.¹³

2.3 Focus

The concept of operations focus was first highlighted by Skinner in the 1974 *Harvard Business Review* article "The Focused Factory," in which he explained that companies can improve performance by making trade-offs that limit their range of activities.¹⁴ This

principle applies to all functions, including strategy, marketing, and operations. Harvard Business School professor Michael Porter writes in his 1996 article “What Is Strategy?” that “[s]trategy is making trade-offs in competing. The essence of strategy is choosing what not to do.”¹⁵ In setting their strategy, firms must focus on specific activities to perform while explicitly identifying others to avoid.

As it applies to operations, focus was originally defined as making explicit trade-offs between achieving low costs and providing various types of differentiation. In this view, operations must make trade-offs among performance objectives to best align with the business strategy. As Skinner maintained, “[f]ocused manufacturing must be derived from an explicitly defined corporate strategy . . . the choice of focus cannot be made independently by the production people.”

2.3.1 Generic Operations Performance Objectives

What performance objectives might operations managers be trading off? Skinner and others have identified four: cost, quality, delivery, and flexibility.

1. **Cost.** Companies looking to achieve the strategic position of cost leadership must be able to provide products and services at lower cost than their competitors. The operations function can achieve this in two ways. Achieving low unit costs allows companies to profit while offering customers prices below those of their competitors. Alternatively, companies that maintain low levels of capital investment, either through reduced investment in fixed assets or decreased working capital, can offer lower prices by accepting lower profit margins.
2. **Quality.** Strategic differentiation based on higher quality requires that the operations function meet a quality performance objective. Higher quality can be attained by improvements in performance, features, dependability, consistency, or conformance to specifications. An increase in quality often requires a direct trade-off with cost. For example, increasing the thickness of a plastic chair improves stability but uses more plastic resin.
3. **Delivery.** Competing through differentiation in the time it takes to respond to customers’ orders necessitates an operating system that focuses on delivery. Performance objectives of this type can be either reliable delivery or quick response. For product-based companies, quick deliveries can be achieved by maintaining high inventory levels, which clearly comes with a trade-off of higher cost.
4. **Flexibility.** Flexibility can take many forms, each of which serves as a performance objective for a specific type of competitive position. Some firms differentiate their offering by allowing for variations in the timing of delivery or the order size, and the corresponding operations performance objective is flexibility to adjust time to delivery or volume. Companies looking to provide innovative products and services call for an operating system that provides the flexibility to develop and introduce new offerings quickly. If the strategy calls for a wide product offering, operations must be able to supply many different products or services. If the competitive

position calls for customer-specific customization, the operations function must excel in its flexibility to alter outputs accordingly.

2.3.2 Focus Is Restricting the Range of Activities

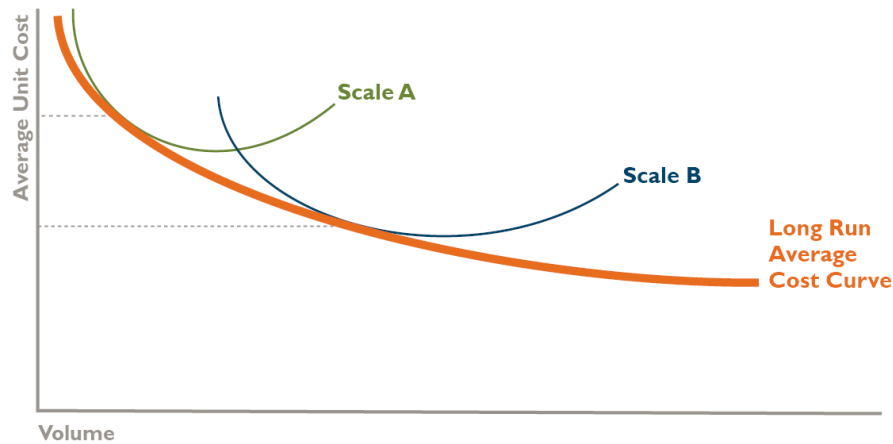
A more general definition of operations focus is any restrictions that limit the range of activities performed. These restrictions can be based on the performance objectives listed above, customer segmentation, geography, technology, process, knowledge, or any other criteria that set boundaries for the operations function. As a general rule, the narrower the focus, the greater operations' ability to excel in the targeted activities.

2.3.3 The Benefit of Focus

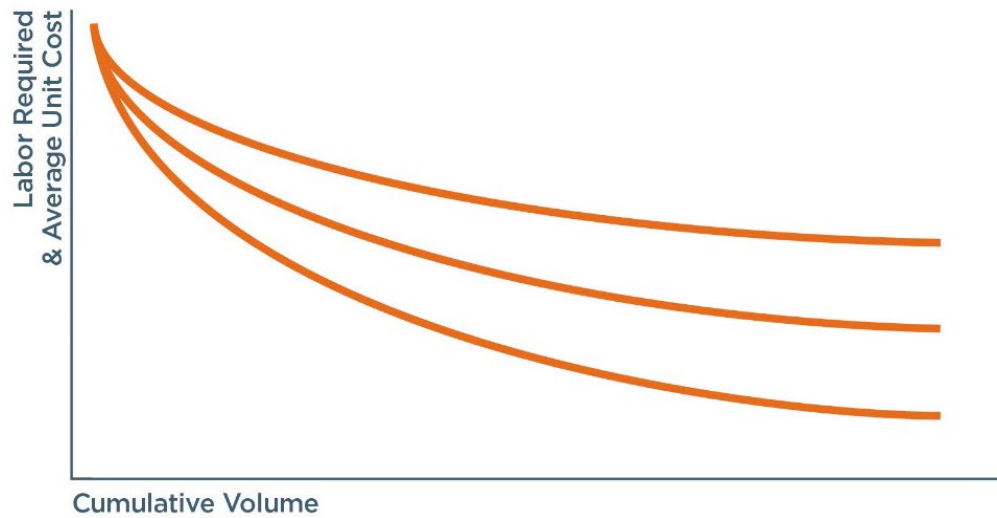
Focus allows for *operational specialization*, which increases operational performance in several ways. First, it reduces complexity. Management attention is directed only to certain activities, which greatly simplifies decision making. And because the organization is performing fewer activities, its complexity of coordination is greatly reduced.

Second, operational specialization contributes to efficiency. Physical and knowledge-based resources can be optimized for the area of focus because "equipment, supporting systems, and procedures can concentrate on a limited task," according to Skinner. Efficiencies improve as operations employs specialized resources selected to best meet performance objectives in the area of focus.

A focused operation is better suited to benefit from efficiencies gained through ***economies of scale***, decreasing average costs as volumes increase (see **Exhibit 4**). Increased volumes allow the company to use physical and human resources more efficiently, employ alternative process technologies, and increase negotiating power with suppliers, all of which can reduce average cost per unit.

EXHIBIT 4 Economies of Scale

Third, specialization leads to learning. As Skinner pointed out, “simplicity, repetition, experience, and homogeneity of tasks breed competence.”¹⁶ By performing only a small set of activities, the company learns how to do them better. The **learning curve**, first identified in the manufacturing of airplanes in the 1930s, is the observation that the amount of time (and therefore costs) required to perform a given task decreases with experience, as shown in **Exhibit 5**. Learning is often referred to as *dynamic economies of scale* because it relates decreasing average costs to increases in *cumulative* volume. Employees become more valuable as they learn to perform repeated tasks more efficiently, gaining expertise in their focused set of activities. In addition, the overall workforce becomes more efficient as individual employees learn how best to interact with one another in producing outputs.

EXHIBIT 5 Learning Curve

Source: Based on learning curve concept developed by T. P. Wright, "Factors Affecting the Cost of Airplanes," *Journal of Aeronautical Sciences*, 3(4) (1936): 122–128 and extended in Bruce Henderson, "The Experience Curve," *BCG Perspectives*, Boston Consulting Group, 1968.

2.3.4 Loss of Focus

A company can lose focus for several reasons. Sometimes, additional activities are slowly added over time, a phenomenon known as *drift*. In other cases, companies experience what is known as the *focus paradox*, in which the efficiencies and success gained through operational specialization lead a company to explore avenues for growth. At some point, however, growth can be achieved only by undertaking activities outside the range of focus, resulting in a loss of focus. Complex strategies often call for the operations function to take on a wide range of activities. The benefits of focus and operational specialization can still be achieved by splitting the activities among several units, each of which focuses on a specific subset. The independent-focused operations form a network that collectively completes a wide range of activities and enables the complex strategy.

2.4 The Operations Strategy Framework

The operations strategy framework is a tool for gauging the strategic alignment of the operations function. It helps determine whether operations will be able to accomplish its strategic goals and therefore the organization's strategic goal.

The first operations strategy framework was introduced by Skinner in 1969.¹⁷ It was further developed by researchers including Wheelwright and Hayes, who developed a list of decision areas—important choices that constitute the main responsibilities of the "manufacturing division."¹⁸ By reviewing the choices made in key decision areas, they argued, managers could check that operations' actions aligned with the company's overall

strategy and goals. This framework proved very useful for many companies for a couple of decades.

But the original decision areas do not fully capture the important choices of the operations function in a changing business landscape. Service and knowledge-based businesses face different trade-offs than manufacturers do. Companies are much more reliant on the ecosystem of partners, suppliers, and customers that work together to deliver value, and the boundaries of the operations function are not defined by the four walls of the factory.

Over the decades, many scholars have modified the list of decision areas to be relevant to an increasingly diverse set of businesses. A decision-area-based framework will always be limited, however, by the choices that have already been identified, and it might not capture the priorities of the business models of tomorrow. As we will see later in this reading, decision areas have a place in an operations strategy. But the business realities of the modern economy require a more robust operations strategy framework.

A robust framework must meet three basic criteria. First, it must be comprehensive, considering all potentially relevant decisions and actions. Second, it must be generalizable to nearly all kinds of companies—manufacturers, service providers, R&D houses, software developers, or Internet startups. Finally, it must be useful to managers in assessing the strategic alignment of the operations function.

2.4.1 Resource-Based Operations Strategy Framework

Building on the resource-based view of the firm developed by researchers including Edith Penrose and Birger Wernerfelt, we propose a framework that focuses on how operations develop, organize, and utilize resources.¹⁹ The *resource-based operations strategy framework* is used to evaluate strategic alignment by examining the sets of resources employed by the operations function to meet performance objectives.

For this framework to be comprehensive and effective, it must consider all resources that influence the firm's ability to create and deliver its primary offerings, even resources that lie outside the boundaries of the operations function as defined by the company's organizational structure. There is no rule guiding how resources should be considered, but generally the list includes those related to production, warehousing, product R&D, process development, engineering, purchasing, personnel, and finance.

To demonstrate the use of a resource-based operations strategy framework, consider the discount retail group Aldi and its expansion into markets in Europe, the United States, and Australia.²⁰ Aldi thrived with its strategy of offering the lowest prices on groceries and basic goods. By 2012, the group operated more than 9,000 stores in 18 countries; the stores generated €50 billion in sales per year. In this expansion, Aldi's operating system focused on efficiencies in both its retail operations and its products: stores carried less

than 1,500 different items (compared to typical US supermarkets that carried more than 25,000), and about 90 percent were branded under the chain's private labels. To offer extremely low prices and still earn a profit, the operations function must keep costs to a minimum, seeking every efficiency available while maintaining an assortment and quality of products that keep customers coming back.

To draw a complete picture of an operating system, resources can be classified into five distinct categories: physical resources; human resources; intellectual property (IP), software, and methods; ecosystem resources; and financial resources.

1. *Physical resources* are the tangible property, plant, and equipment of the operations function. These include facilities, offices, warehouses, land, machinery, equipment, and tools. Some physical resources—for example, manufacturing facilities and distribution centers—may reflect large investments that are made only once every 20 or more years. Others, like disposable tools, may require little investment and can be changed frequently.

As it expanded, Aldi's operating system included many types of physical resources. Many large distribution centers and thousands of modestly sized stores were organized in hub-and-spoke patterns across 18 countries. Within the distribution centers, the company employed equipment that enabled the efficient flow of inventory. The no-frills stores, each of which contained a small storage room, were equipped with basic shelving arranged in just three or four aisles and little decoration. Shopping carts were available but were equipped with a locking mechanism that required customers to provide a small deposit. Two or three checkout counters with barcode scanners and cash registers were used to complete sales.

2. *Human resources* are the employees that make up the operations' workforce. Individual employees differ in their skills, knowledge, training, ability, flexibility, and terms of employment. They also differ in the role they play.

Staffers at an Aldi store were kept to a minimum. The manager and two or three assistants all participated in each of the activities that must be performed in the store, including stocking shelves, cleaning floors, and operating registers. They were highly flexible and efficient, and in return they earned twice as much as employees of conventional supermarkets.

3. *Intellectual property (IP), software, and methods* are the operations function's knowledge-based resources. They are the collective knowledge held by the organization, not by individual employees. Included in this category are patented and proprietary designs, trade secrets, software, organizational systems, processes, techniques, and information. Google's search algorithm and Microsoft's source code for Windows are two examples of this very valuable class of resource.

These resources have often been classified as intangible, but they are in some sense very tangible. Even though they are not themselves physical, they can be represented in hard copy. Knowledge and processes can be drawn on paper, and software code can be viewed on a computer screen.

Discount retailing is typically thought of as a "low knowledge" business, but some of Aldi's operational success can be attributed to its IP, software, and methods.

Aldi's automated system for ordering and replenishment allowed it to maintain appropriate levels of inventory throughout the supply chain. Later investment included digital initiatives for customer online ordering, including Aldi Online (for nonfood items) and store ordering with curbside pickup, and greater automation to streamline supplier processes. And the company leveraged knowledge from operating a successful, large-scale, discount retail system in Germany as it expanded to other countries.

4. *Ecosystem resources* are the relationships with the suppliers, distributors, customers, and partners that collectively constitute a company's business ecosystem. These influential resources are beyond the direct control of the company. The relationships with and capabilities of other ecosystem members influence both the activities of the operations function and the achievement of its performance objectives.

Aldi relied on suppliers to contribute to achieving both low cost and reasonably high product quality, and so it invested heavily in maintaining and developing its global network of suppliers that could meet those criteria. To save time and labor, Aldi required suppliers to deliver products on standard-size pallets, use containers that also serve as display cases, and place multiple barcodes on each package to make swiping easier at the checkout counter. Aldi conducted daily checks on quality. Even though prices were generally 20 percent to 30 percent less than those at standard supermarkets, many Aldi products have earned higher quality ratings than leading national brands in customer surveys.

Aldi also expected customers to take an active role in maintaining the low-cost structure. Customers bagged their own groceries and returned shopping carts to designated areas. By transferring some activities to customers, Aldi was able to staff each store with fewer employees.

5. *Financial resources* are the types of funding available, including cash on hand, cash generated from operations, and funds available from financial markets. These resources are generally not considered part of operations, yet they can be decisive in enabling or restricting the implementation of the function's strategy. The ability of operations to develop or realign other types of resources is highly dependent on the financial resources available. For example, a \$500 million investment in automation may be financially and strategically prudent because it can improve efficiency and reduce costs, but it will not happen if operations does not have the funds to invest.

Having been successful for many years, Aldi had sufficient access to financing to invest both in improving and expanding its operations. In 2012, the Aldi group announced that it would invest €3 billion to modernize or replace many of its stores. In 2018, Aldi would have more than 1,600 stores in the United States. By 2022, that number had grown to 2,000 stores, and Aldi had expanded in Asia, with more than 25 stores in Shanghai alone.²¹

2.4.2 The Operations Strategy Matrix

An operations strategy can best be captured in the form of a matrix (see **Exhibit 6**). The columns of the matrix represent the categories of resources (such as the categories already

discussed in this reading) that influence the firm's ability to create and deliver products and services. The rows represent decisions relating to the stock of available resources, the connections (integration) among them, and the development and acquisition of new resources.

1. Resource base is an inventory of the resources of the operations function that captures the cumulative decisions about which resources to employ. The type, location, quality, durability, capacity, and utilization of each resource are listed in this row of the matrix.
2. Resource integration decisions determine how individual resources interact with one another to form the operating system, including decisions about organization, planning, and decision making. Interactions occur both within and across resource categories.
3. Resource development and acquisition decisions guide the development of the operating system. The plans and processes for innovation, development, and mergers and acquisitions determine how the resource base and resource integration are expected to change.

EXHIBIT 6 Aldi's Resource-Based Operations Strategy Matrix (2012)

	Physical Resources	Human Resources	IP, Software, and Methods	Ecosystem Resources	Financial Resources
Resource Base (Type, Capacity, and Utilization)	-Hundreds of midsize warehouses -Thousands of small, no-frills stores -Equipment in store—carts, registers, shelving	-Efficient and motivated store staff -Ability to complete many types of work	Automated inventory management system Experience in hub-and-spoke supply chains	Global network of suppliers Customers participate in cost reductions	Deep pockets of owners
Resource Integration (Organization, Planning, and Decision Making)	-18 countries in Europe, the United States, and Australia -Hub-and-spoke system of 50 to 80 stores Five to six weekly deliveries	Three to five employees per store perform all tasks	Automated system allows frequent deliveries and reduces the number of employees required	-Package serves as display -Lots of supplier quality checks Customers bag groceries and return carts	-Operations is focus of investments Extensive funds required to expand network
Resource Development and Acquisition (Innovation, Development, and M&A)	-Replication of hub-and-spoke system Some M&A, mostly organic expansion	-High salary attracts and retains employees	Refine processes in hub-and-spoke system and roll out to all	Long process to become supplier Maintain strong relationships with current suppliers	Profits used to fund expansion and improvements

Sources: Adapted from Marc Sachon, "Aldi: Disruptor Disrupted?" Case IES944 (Barcelona: IESE Publishing, 2023). Adapted by permission of *Harvard Business Review* from Jan-Benedict Steenkamp and Nirmalya Kumar, "Don't Be Undersold!," *Harvard Business Review* 87, no. 12 (2009). © 2009 by the Harvard Business School Publishing; all rights reserved.

2.4.3 Assessing the Operations Strategy

Using the matrix to lay out the operations strategy allows for the evaluation of the complete set of influential decisions made by operations. An operations strategy can be evaluated on two dimensions.

First, it must be internally consistent—that is, decisions relating to one kind of resource must complement decisions about the other resources with which it interacts. For example, employing a process technology that simplifies the nature of tasks on an assembly line is consistent with the decision not to staff the line with highly trained engineers.

Second, an operations strategy must be strategically aligned with the business strategy so that the operating system is able to meet all of operations' performance objectives derived from the business strategy. The contribution of the operations strategy to future competitive advantage can also be gauged through the development and strengthening of resources that differentiate the company from competitors in a way that positions the company for future success.

The most basic measure of **strategic alignment**, sometimes called *fit*, is *simple consistency* between each resource-related decision and the company's strategic priorities. Because operations acts through systems of resources that interact with one another to create and deliver outputs, a more sophisticated analysis of strategic alignment must consider those interactions. They can reinforce one another, creating a level of alignment that is more than the sum of their simple consistencies. In some cases, systems of resources can best meet performance objectives when some of the resources do not achieve simple consistency; the strategic alignment of the entire system may be *optimized*, the highest level of alignment, because individual elements are not.

In addition to demonstrating simple consistency, Aldi's decisions related to physical resources are reinforcing, allowing stores to avoid stock-outs while keeping inventory low. The facilities are laid out in an efficient hub-and-spoke pattern, in which each distribution center supplies 50 to 80 stores, none of which is more than 50 kilometers away. Each store is relatively small and contains a small storage area, saving on space and its associated costs. Because stores are very close to the distribution center, deliveries are made five or six times per week, keeping the shelves stocked even though there is little storage and shelf space.

Employees in Aldi stores are paid double what competitors offer, which, on the surface, seems inconsistent with Aldi's low-cost strategy. But this decision is just one in an optimized system. Human resources interact with IP, software, and methods; ecosystem resources; and physical resources in a way that is strategically aligned. The automated ordering system simplifies ordering, suppliers simplify stocking by using pallets of standard size and containers that transform easily into display cases, customers simplify the checkout process by bagging their own groceries, and store design simplifies cleanup. Because of all these decisions, a typical Aldi store can be adequately staffed with only three or four flexible and efficient employees who all participate in every aspect of operating the store. The efficiencies gained by Aldi's more-skilled employees justify their higher wages. Overall, the interaction of this human resources decision with Aldi's other decisions produces a significant cost advantage for the company. Each store's workforce costs Aldi only three percent of sales, significantly lower than the nine percent for a standard supermarket.

2.4.4 Operations as a Complex System

An operating system is a complex collection of many types of resources. Even though the resource-based framework highlights the individual components, they act as a single system. Creating it is much more difficult than just assembling the components. Even if competitors can copy all the parts, they will find it very difficult to replicate their integration successfully.

Because of its complexity, however, an operating system can become rigid. The organization and interaction of resources may become so ingrained that the system can become inflexible and lose its ability to adapt.

The collection of resources in an operating system is continuously evolving—even if the system through which the resources operate becomes rigid—and so the operations strategy must be monitored and evaluated regularly. Changes to only a few elements can have a large impact overall. For example, a manufacturer in Germany looking to open a plant in Thailand may initially consider replicating its current facility and equipment. Because wages are significantly lower in Thailand, however, the optimal setup may include many more workers and fewer productivity tools. In this case, one operational change—the location of the facility—necessitates many other changes.

2.5 Operations Decision Areas

The resource-based framework we have been discussing is one way to assess the completeness and alignment of an organization's operational resources. As we have noted, it was preceded by other frameworks. The first, developed by Skinner in 1969, focused on manufacturing operations. Skinner identified what he called “some important trade-off decisions in manufacturing”—major operations *decision areas* where management's choices can have a large impact on the deployment of operating resources and thus on operating performance (see **Exhibit 7**). This framework examined strategic alignment by assessing whether choices in each decision area were consistent with the competitive priorities of the operations function and of the company as a whole. These decision areas can be categorized as either *structural* or *infrastructural*.

Structural decisions can be thought of as the “hardware” of operations. They are infrequent, require substantial financial investments, and are difficult to make and to reverse. Thus, they affect the structure of an organization for many years. Structural decision areas include:

1. **Capacity.** The amount of output that the operating system can reliably deliver. Capacity decisions relate to each type of product or service provided, the flexibility to increase or decrease volumes to meet changes in demand, and the timing of capacity additions or subtractions.

2. **Facilities.** The location, size, and specialization of the physical facilities of the operations function. The network of facilities must be developed and coordinated so that each facility understands the types of output it produces and the markets it serves.
3. **Scope.** The level of vertical integration, supply chain position, and relationships with suppliers. The scope of the operations function is tightly linked to the company's market position defined by its strategy.
4. **Information and process technology.** The technologies used in the operating system. Choices include the degree of capital or labor intensity, the design of the production system, the level of automation, and the process technology. These decisions can have a strong impact on the company's competitive positioning. For example, specialization may allow output to be produced at lower costs but restricts flexibility in responding to market changes.

Infrastructural decisions are the many small choices that create the “software” that complements the so-called hardware. Each of these decisions is relatively easy to make, low in cost and impact, and reversible to some degree; cumulatively, however, they can restrict or enable the operations function's ability to meet its performance objectives. Infrastructural decision areas include:

1. **Workforce.** The types and skill levels of employees, the way they are selected and trained, the way their performance is measured, and the way they are motivated. These decisions can have a tremendous impact on both the company's cost structure and the type of output it can produce.
2. **Organization.** The organizational structure, roles and responsibilities, and degree of centralization. Organization decisions define who will be responsible for making and implementing many of the key strategic choices made by the operations function.
3. **Quality.** Systems to ensure that output meets desired quality levels. The appropriate quality measurement and control system will depend on decisions relating to process technologies and the workforce.
4. **Production planning and distribution.** Production scheduling, inventory management, warehousing, and delivering output to customers.
5. **Product and process development.** Investments in innovation, responses to external developments, responsiveness to customer requests for additional or improved offerings, and techniques for learning from experience.

EXHIBIT 7 Operations Decision Areas

STRUCTURE	INFRASTRUCTURE
Capacity Amount Timing Type	Workforce Skill level Wage policies Employment security
Facilities Size Location Specialization	Organization Structure Control / reward system Role of staff groups
Scope Direction Extent Balance	Quality Defect prevention Monitoring Intervention
Information and Process Technology Equipment Automation Linkages	Production Planning and Distribution Production scheduling Sourcing and inventory management Warehousing and delivering
	Product and Process Development Investments in innovation Responses to development of others Learning from experience

Source: Adapted from Steven Wheelwright, "Japan—Where Operations Really Are Strategic," *Harvard Business Review* 59, no. 4, (1981): 68.

Because the decision areas were originally developed for a manufacturing setting, the terminology may not be intuitive in other situations. Yet managers in all sorts of industries face many of the same decisions. Mobile phone operator China Mobile must measure the quality of its service, video game developer Electronic Arts must make decisions about its capacity for developing games, and semiconductor designer ARM must assess its scope of operations. The decision areas framework has been adapted again and again to better fit service, research, and software companies in a wide range of industries. In these instances, the plant network may become a facilities network, and production planning may become R&D portfolio strategy, but the basic structure of the framework is similar. In essence, the decision area framework captures most of the important resource-based decisions that affect an organization's operating performance.

Coupled with the resource-based operations strategy framework, these decision areas can serve as a guide for assessing the strategic alignment of an operations strategy. There are many ways to analyze an operations strategy. The best framework lets managers easily and accurately assess strategic alignment between operations decisions and business strategy, enabling an honest discussion of the operating systems and its ability to meet performance objectives.

3 SUPPLEMENTAL READING

3.1 Managing Disruptive Strategy Transitions: Microsoft Server and Tools

In 2012, Microsoft was the world's largest software company, with sales of \$74 billion from products, including the extremely popular Windows and Office software for personal computers.²² About one-quarter of Microsoft's revenues came from the Server & Tools Business (STB), the division that focused on developing software for the servers that underpin corporate computer networks. Like the Windows and Office divisions, STB developed sophisticated software products that could be used in large numbers of applications and released major versions of each product every three to five years. The division enjoyed a very successful business model, with strong alignment around its operating strategy. Then the disruptive emergence of cloud computing began to threaten its success.²³

3.1.1 STB's Traditional Operations Strategy

STB's traditional operations strategy worked for many years to deliver server software products (such as Windows Server and SQL Server) on regular cadences, or product-development cycles. New versions were released every few years, expanding feature sets and driving a lower total cost of ownership than that of competing systems such as Novell and LINUX. The software was sold as a packaged product to corporate IT departments (see **Exhibit 8**).

1. *Physical resources.* STB's traditional business model required low investment in physical resources. Investments were limited to a variety of IT equipment and office facilities in Microsoft's Redmond, Washington, campus.
2. *Human resources.* Highly skilled and well-paid software engineers and project managers designed, developed, tested, and coordinated STB's software products. Most of the division's 9,000 employees were dedicated to product development. Because Microsoft risked losing talented

employees to startups and competitors, it offered top salary and financial incentives to attract and keep top talent.

3. *IP, software, and methods.* IP is essential to Microsoft's operations strategy. Central to this is the product's code base, which has been developed, tested, used, and reused by thousands of software engineers. STB's processes and methods are valuable assets.
4. *Ecosystem resources.* STB provided only one component of the solution desired by corporate IT departments. Hardware vendors produced the physical servers, other software developers produced applications, system integrators built entire systems, and corporate IT departments and data center operators implemented and operated the servers and software. As part of its operations strategy, Microsoft reached out to its ecosystem of service and hardware providers through investments, partnerships, conferences, and programs to make sure they were up to speed and aligned on the newest software, strategies, and methods. Overall, Microsoft invested billions of dollars in ecosystem-oriented technologies and in relationships with more than 30,000 partners.
5. *Financial resources.* Developing sophisticated software can be very expensive. Each major release of Windows Server, for example, cost over \$1 billion—mostly in human resources—to develop. STB spent more than \$2 billion annually on product development, financing this sum through the profits generated from existing products. Lucky for Microsoft, it has been in a cash-rich position during most of its history.

EXHIBIT 8 Microsoft STB's Traditional Operations Strategy

	Physical Resources	Human Resources	IP, Software, and Methods	Ecosystem Resources	Financial Resources
Resource Base (Type, Capacity, and Utilization)	Standard computers and network equipment Generic offices	Over 8,000 highly skilled and talented software engineers Project managers	Existing code base Process to coordinate development by thousands of developers	Only part of solution for end customer Application developers build on top of platform	Profits from existing product lines
Resource Integration (Organization, Planning, and Decision Making)	Maintain sufficient capacity to not inhibit software development	Activity coordinated to ensure progress according to development plan	Source code maintained centrally Fixed development process	Work with many third parties in software, hardware, and system integration	Development costs of \$1B per release over 2–3 years before revenues realized

Resource Development and Acquisition (Innovation, Development, and M&A)	Continue use of off-the-shelf products	Top salary and incentives to attract and retain talent	Continually code new features	Continue to promote platform to attract new software developers and partners	Large installed base
	Increase along with head count	Recruit from universities	Acquire other software to add to code base		50% of revenue from annual licenses

Source: Adapted from Marco Iansiti and Alain Serels, "Microsoft Server & Tools," HBS No. 613-031 (Boston: Harvard Business School, 2013).

3.1.2 A Disruptive Change in STB Strategy

In 2010, Microsoft recognized the growth of cloud computing as a key trend. Cloud-based competitors such as Amazon Web Services (AWS) began to present a disruptive threat to Microsoft's business model by offering hosted copies of LINUX delivered as a service. Although the performance and features were initially perceived as inferior (less secure and less responsive, for example) the convenience provided an incentive for the lower tiers of the market to adopt the software. Microsoft's fundamental business model was threatened, and the company's STB was one of the main battlegrounds. While its traditional business was still growing at around 10 percent per year, with impressive profit, STB's leadership shifted its strategic focus to encompass cloud-based versions of the server products. Instead of shipping only packaged software, STB was also to sell the service of using the software.

The emergence of cloud computing represents one of the most dramatic transitions in the history of computing: Software companies traditionally dominated by development and sales operations now must build extensive operations infrastructure to deliver the software as a service. The deployment and provisioning of the data centers that serve the software on demand, for example, require profoundly different capabilities. Cloud service operations also require sophisticated supply chain capabilities and leverage a completely different, pay-by-usage business model.

In addition to building and operating the hardware infrastructure, Microsoft's STB was transforming itself from a pure software developer to a service provider. Customers relied on Microsoft's cloud every day to run their businesses. Previously, STB only had to develop the software and hand it off to the server operator. Microsoft must guarantee that the cloud service, delivered by operating software and hardware, has 99.999 percent reliable uptime. And it must continue to operate its traditional business, backed by its traditional operations strategy: shipping license-based packaged software.

Operating a cloud computing service requires major changes across every element of the operations strategy framework. By early 2013, STB's transition was still in the early stages, but much progress was already apparent (see **Exhibit 9**).

1. Physical resources. Being a cloud service provider is extremely asset-intensive. STB had been building a global network of massive state-of-the-art data centers and equipping them with hundreds of thousands of servers. In 2013, STB anticipated \$1.2 billion in capital expenditures. These massive expenditures are naturally coupled with dramatic changes in STB's capability base and operating requirements. Rather than only shipping software DVDs in a box, the business now also operates a continuous, highly reliable service, which can ramp up and ramp down capacity on demand.
2. Human resources. The technical diversity involved in operating the cloud service is dramatically greater. In addition to development teams, STB needed engineers to plan, operate, and maintain the data centers. It also needed customer support teams to respond to customer problems in real time.
3. IP, software, and methods. STB's existing software was a key resource in delivering cloud services. But the business had to modify and, in some cases, completely rewrite the packaged software so that it was optimized for running on a large-scale cloud network. The company needed to build knowledge and procedures for establishing and running efficient large-scale data centers—something that very few companies had accomplished.
4. Ecosystem resources. STB began taking on a much larger role in delivering the overall solution for customers, now providing the data center, hardware, Windows Server operating system, and applications such as SQL Server. In doing so, it created opportunities for third-party application developers, who can offer cloud-based applications more easily by building on top of STB's cloud platform. The ecosystem is shifting, and Microsoft's programs and partnerships shifted with it.
5. Financial resources. Building the global network of data centers has cost several billion dollars, much of which had to be spent before any sales could be made. At the same time, the cloud business model has shifted many customers from making large upfront purchases of packaged software to smaller monthly payments based on cloud usage. For those two reasons, the transformation has required tremendous sums of capital. STB has been able to finance much of this through the continued strong sales of its packaged products.

EXHIBIT 9 Microsoft STB's Cloud Operations Strategy (2013)

	Physical Resources	Human Resources	IP, Software, and Methods	Ecosystem Resources	Financial Resources
Resource Base (Type, Capacity, and Utilization)	Multiple large data centers equipped with specially designed servers	Software developers Data center designers and operators Customer service teams	Existing code base Knowledge of data center design and operation	Offer full cloud solution Platform for application developers to offer cloud products	Profits from existing packaged products
Resource Integration (Organization, Planning, and Decision Making)	Global network of data centers Very high utilization 99.999% uptime	Immediate response to outage and/or error Small teams continuously improve software	Software is just part of offering Data center software to maintain high utilization	Rely less on hardware and integrators Application developers become customers	Large upfront investment in cloud model Invest over \$1 billion per year in data center development
Resource Development and Acquisition (Innovation, Development, and M&A)	Build and/or expand data center network Add server capacity ahead of demand	Top salary and incentives to attract and retain talent Recruit from other cloud providers	Rewrite and/or add to code base to optimize for cloud Develop data center operations knowledge	Promote platform as entry for application developers to cloud	Need to build scale in cloud to recoup high fixed cost from subscription model

Source: Adapted from Marco Iansiti and Alain Serels, "Microsoft Server & Tools," HBS No. 613-031 (Boston: Harvard Business School, 2013).

3.1.3 Results

By 2013, Microsoft's STB effectively operated as two related organizations, each with its own operations strategy. Both reported to Satya Nadella, then STB president. The resource-based operations strategy framework provided a useful way to examine the range of changes necessary to implement the new strategy and rapidly generate a list of focus categories and actions that will either impede or enable the transformation. The framework became a useful tool in understanding the different priorities of the two sides of the business as well as the different capabilities and investments involved.

Microsoft continued to operate the two businesses side by side during the transition from packaged software to cloud computing. The traditional business was expected to continue for at least another decade, so this ambidextrous phase (in which STB operates effectively as two related organizations) will be a long one. By the spring of 2013, Microsoft

had made significant progress toward this goal, and STB's cloud business reached \$1 billion in revenue.

4 KEY TERMS

business unit strategy Strategic plan of a business unit within a corporation to compete in its market.

corporate strategy Strategic plan of the corporation, including the selection of markets, organizational structure, and resource allocations.

deliberate strategy Portion of the directed strategy that is actually implemented.

directed strategy Strategic plan for the development and organization of resources to create and capture value.

economies of scale Reduction in average cost per unit as total number of units being produced increases.

emergent strategy Actual choices made that were not part of the directed strategy.

functional strategy Strategic plan of a function within a business unit to implement the business unit strategy.

learning curve Increased efficiency and reduction in average cost per unit gained from experience.

operations strategy A framework or plan for the utilization of resources that drive business execution.

realized strategy The actual choices made that delineate a company's activities and define how it creates and captures value.

strategic alignment Coherence between actual choices made and the objectives of the company's strategy. Also known as *fit*.

5 FOR FURTHER READING

Batato, Magdi, Xavier Mesnard, and Suketu Gandhi. "It's Time for a New Model for Operations Management." *Harvard Business Review*, September 19, 2023. <https://hbr.org/2023/09/its-time-for-a-new-model-for-operations-management>. Accessed April 2024.

Iansiti, Marco. "The History and Future of Operations." *Harvard Business Review*, June 30, 2015 (HBS No. H026PQ).

Porter, Michael. "What Is Strategy?" *Harvard Business Review* 74, no. 6 (1996): 61–78.

Reeves, Martin, and Adam Job. "Operations in an Era of Radical Uncertainty." *Harvard Business Review*, October 2, 2023. <https://hbr.org/2023/10/operations-in-an-era-of-radical-uncertainty/>. Accessed April 2024.

Skinner, C. Wickham. "The Focused Factory." *Harvard Business Review* 52, no. 3 (1974): 113–121.

Skinner, C. Wickham. "Manufacturing—Missing Link in Corporate Strategy." *Harvard Business Review* 47, no. 3 (1969): 136–145.

6 ENDNOTES

¹ Dictionary.com Dictionary, s.v. “operation,” accessed May, 2024, <https://www.dictionary.com/browse/operation>.

² World Bank, “Services, Value Added (% of GDP)—United States,” 1997–2021, <https://data.worldbank.org/indicator/NV.SRV.TOTL.ZS?locations=US>, accessed April 2024.

³ See World Trade Report 2019: *The Future of Services Trade*, https://www.wto.org/english/res_e/publications_e/wtr19_e.htm, accessed April 2024; World Bank, World Development Indicators, “Services, Value Added (% of GDP)—China,” 1960–2022, <https://data.worldbank.org/indicator/NV.SRV.TOTL.ZS?locations=CN>, accessed April 2024.

⁴ Karim R. Lakhani, Yael Grushka-Cockayne, Jin Hyun Paik, and Steven Randazzo, “Customer-Centric Design with Artificial Intelligence: Commonwealth Bank,” HBS No. 622-065 (Boston: Harvard Business School Publishing, 2021). See also Andy Wu, Matt Higgins, Miaomiao Zhang, and Hang Jiang, “AI Wars,” HBS No. 723-434 (Boston: Harvard Business School Publishing, 2023); Marco Iansiti and Karim R. Lakhani, *Competing in the Age of AI: Strategy and Leadership When Algorithms and Networks Run the World* (Boston: Harvard Business Review Press, 2020).

⁵ Wickham Skinner, “Manufacturing—Missing Link in Corporate Strategy,” *Harvard Business Review* 47, no. 3 (1969): 136–145.

⁶ Wickham Skinner, “Manufacturing—Missing Link in Corporate Strategy,” *Harvard Business Review* 47, no. 3 (1969): 136–145.

⁷ Wickham Skinner, “Manufacturing—Missing Link in Corporate Strategy,” *Harvard Business Review* 47, no. 3 (1969): 136–145.

⁸ See Novartis, *Annual Report 2020*, p. 52, <https://www.novartis.com/sites/novartis.com/files/novartis-annual-report-2020.pdf>, accessed April 2024.

⁹ This definition, long taught to first-year Harvard Business School MBA students, is based on the work of Michael Porter, “What Is Strategy?,” *Harvard Business Review* 74, no. 6 (1996): 61–78.

¹⁰ The Walt Disney Company, “Disney CEO Bob Iger Delivers Message of ‘Renewed Strength’ at Annual Shareholders Meeting,” April 3, 2024, <https://thewaltdisneycompany.com/disney-shareholders-bob-iger-remarks/>, accessed May 2024. See also The Walt Disney Company, Fiscal Year 2023 Annual Financial Report, <https://thewaltdisneycompany.com/app/uploads/2024/02/2023-Annual-Report.pdf>, accessed May 2024.

¹¹ Dollar General, *2020 Annual Report and Proxy Statements*, <https://investor.dollargeneral.com/websites/dollargeneral/English/3200/annual-reports.html>, accessed April 2024.

¹² Wickham Skinner, “Manufacturing—Missing Link in Corporate Strategy,” *Harvard Business Review* 47, no. 3 (1969): 136–145.

¹³ Robert Hayes and Steven Wheelwright, *Restoring Our Competitive Edge: Competing Through Manufacturing* (New York: John Wiley, 1984), 35–42.

¹⁴ Wickham Skinner, “The Focused Factory,” *Harvard Business Review* 52, no. 3 (1974): 113–121.

¹⁵ Michael Porter, “What Is Strategy?,” *Harvard Business Review* 74, no. 6 (1996): 61–78.

¹⁶ Wickham Skinner, “The Focused Factory,” *Harvard Business Review* 52, no. 3 (1974): 115.

¹⁷ Wickham Skinner, “Manufacturing—Missing Link in Corporate Strategy,” *Harvard Business Review* 47, no. 3 (1969): 136–145.

¹⁸ Robert Hayes and Steven Wheelwright, *Restoring Our Competitive Edge: Competing Through Manufacturing* (New York: John Wiley, 1984).

¹⁹ See Edith Penrose, *The Theory of the Growth of the Firm* (Oxford: Blackwell, 1959); and Birger Wernerfelt, “The Resource-Based View of the Firm,” *Strategic Management Journal* 5, no. 2 (1982): 171–180.

²⁰ See, for example, Eric Van Den Steen and David Lane, “Aldi: The Dark Horse Discounter,” HBS No. 714-474 (Boston: Harvard Business School Publishing, 2016).

²¹ See Marc Sachon, “Aldi: Disruptor Disrupted?” (Barcelona: IES Publishing, 2023).

²² This section draws heavily from the main case and the two supporting cases by Marco Iansiti and Alain Serels: Marco Iansiti and Alain Serels, “Microsoft Server & Tools,” HBS No. 613-031 (Boston: Harvard Business School Publishing, 2013); Marco Iansiti and Alain Serels, “Microsoft Server & Tools (B),” HBS No. 613-046 (Boston: Harvard Business School Publishing, 2013); Marco Iansiti and Alain Serels, “Microsoft Server & Tools (C),” HBS No. 613-047 (Boston: Harvard Business School Publishing, 2013).

²³ For a discussion of disruptive technology, see Clayton M. Christensen, *The Innovator’s Dilemma* (Boston: Harvard Business School Press, 1997).

7 INDEX

Page numbers followed by *f* refer to figures. Page numbers followed by *t* refer to tables.

- acquisitions, 18. *See also* resource development and acquisitions
- decisions
- Adaptive Decision Models (ADMs), 5
- AirBnb, 4
- airplane manufacturers, 14, 14*f*
- Aldi, 16–18, 19*t*, 20
- alignment, in operations strategy, 20, 21
- Apple, 5
- application developers, 4, 25*t*, 27, 28*t*
- ARM, 6
- artificial intelligence (AI), 5
- automation, 11, 17, 18, 22, 23*f*
- AWS (Amazon Web Services), 26
- automation, 10, 16, 20, 21*t*

- biotech firms, 5
- Boeing, 4
- business models, 4–5
- business unit strategy, 9, 9*f*, 29

- capacity (decision area), 18, 19*t*, 21, 23*f*
- capital investment, 12
- channel management, 3
- China, 4
- China Mobile, 23
- cloud computing, 24–28
- cloud operations, Microsoft, 28*t*
- communications technologies, 5
- competence, 14
- competition, 7, 9
- competitive advantage, 5, 9, 10, 11, 20
- complexity, 4, 13, 21

- consistency, in operations strategy, 12
- consulting firms, 4
- content of operations strategy, 8
- corporate strategy, 6, 7, 8–9, 9*f*, 12, 29
- cost, as performance objective, 12
- cumulative volume, 14
- customer segmentation, 13

- data as asset, 5
- decision areas, 15, 21–23, 23*f*
- decision making, 6–7, 13, 18, 19*t*, 26*t*, 28*t*
- delivery, as performance objective, 12
- development costs, 6, 25*f*
- differentiation, 3, 5, 12
- directed strategy (deliberate strategy), 8–11, 9*f*, 10*f*, 29
- discount retailing, 17
- distributed operations, 5
- distribution centers, 16, 20
- Dollar General, 10
- drift, 14
- dynamic economies of scale, 14

- economies of scale, 13–14, 13*f*, 29
- ecosystem resources, 16, 17, 19*t*, 20, 25, 25*t*, 27, 28*t*
- efficiency, 5, 8, 11, 13, 14, 16, 18, 19*t*, 20
- electric utilities industries, 5
- Electronic Arts, 23

- emergent strategy, 10–11, 10*f*, 29
- facilities (decision area), 22, 23*f*
- factory management, 4
- factory operations, 4, 6, 11, 15
- finance function, 9
- finance strategy, 8, 9*f*, 11
- financial resources, 3, 5, 16–17, 18, 19*t*, 25, 25*t*, 27, 28*t*
- fit, in operations strategy, 20
- flexibility, as performance
 - objective, 12
- focus, 11–14
- focus paradox, 14
- Ford, 4
- functional strategy, 9, 9*f*, 29
- Genentech, 5
- General Electric, 4
- Genzyme, 5
- global network of suppliers, 17, 19*t*
- global supply chain, 4
- globalization, 5
- goals, 7, 15
- Google, 5, 6, 17
- gross domestic product (GDP), 4
- Hayes, Robert, 11, 15
- health care industries, 5
- human resources, 16–17, 27
- information and process
 - technology (decision area), 22, 23*f*
- information economy, 5
- information resources, 3
- information technology (IT), 5, 24, 25
- infrastructural decisions, 21, 22
- innovation, 18, 22, 23
- intellectual property (IP), 4, 17, 25, 25*t*, 27, 28*t*
- Intellectual Ventures, 4
- inventory flow, 16
- inventory levels, 12, 17
- inventory management, 19*t*, 22, 23*f*
- investment banks, 4
- knowledge-based assets, 5
- knowledge-based business, 4, 5, 15
- knowledge-based resources, 13, 17
- leadership, 4*f*, 12, 26
- learning, 14, 14*f*, 22, 23*f*
- learning curve, 14, 14*f*, 29
- LINUX as a service, 26
- managers, 4, 7, 10–11, 23–24
- manufacturing, 3–5, 7, 11, 12, 15, 16, 21, 23
- manufacturing facilities, 16, 21, 23
- manufacturing outsourcing, 4
- marketing function, 9, 11
- marketing strategy, 8, 9, 9*f*
- Mediatek, 6
- mergers and acquisitions, 18. *See also* resource development and acquisition decisions
- Microsoft, 4, 5, 17, 24–29, 25*t*, 26*t*, 28*t*
- mobile phone manufacturing, 6
- modularization, 5
- Nadella, Satya, 28
- Nike, 4
- Novartis, 8
- objectives, performance
 - See* performance objectives
- objectives of operations, 8
- operating decisions, 6, 11
- operation, definition, 3
- operational specialization, 13–14, 22
- operations decision areas. *See* decision areas

- operations focus, 4, 11–14
- operations managers, 12, 15. *See also* managers
- operations strategy, 4*f*, 6–11, 9*f*
- operations strategy evaluation, 19–20
- operations strategy examples, 21–29, 25*t*, 28*t*
- operations strategy matrix, 18–19, 19*t*
- optimized alignment, in operations strategy, 20
- organization (decision area), 22, 23*t*
- outsourcing, 4, 5
- partner network, 5, 15, 17, 25, 27
- patent portfolio managers, 4
- patents, 4, 5, 17
- Penrose, Edith, 15
- performance improvements, 6, 9, 10, 11–12
- performance objectives, 6, 9, 12–13, 15, 17, 20, 22
- pharmaceutical companies, 8
- physical resources, 16, 19*t*, 20, 24, 25*t*, 27, 28*t*
- plan, for objectives, 8
- Porter, Michael, 8, 11
- product and process development (decision area), 22, 23*f*
- product development, 3, 6, 8, 16, 22, 23*f*, 24, 25, 25*t*, 26
- production management, 4, 22, 23*f*
- production operations, 5–7, 16, 22, 23*f*
- production outsourcing, 4
- production planning and distribution, 22, 23*f*
- production process, 3
- production scheduling, 22, 23*f*
- profitability, 19*t*, 25, 25*t*, 26, 28
- project management, 25
- proprietary technologies, 5, 17
- Qualcomm, 6
- quality (decision area), 12, 22
- quality, as performance objective, 12
- quality control, 10–11
- raw materials, 3
- realized strategy
 - See* emergent strategy
- research and development (R&D), 8, 15, 16, 23
- resource allocation, 9
- resource base decisions, 18, 19*t*, 25*t*, 29*t*
- resource classes (categories), 5, 17
- resource development, 18, 19*t*, 26*t*, 28*t*
- resource development and acquisition decisions, 18, 19*t*, 26*t*, 28*t*
- resource integration decisions, 18, 19*t*, 25*t*, 28*t*
- resource utilization, 10
- resource-based operations strategy, 15–18, 19*t*, 21, 23
- resources, 3–5, 7–8, 9, 11
- retailers, 4, 10, 16, 17
- SaaS providers, 4
- Samsung, 6
- scope (decision area), 22, 23*f*
- search algorithms, 5, 17
- search data, 5
- senior management, 6, 7, 8
- Server & Tools Business (STB), 24–29
- service businesses, 3, 4, 5, 12, 15, 25
- service management, 3
- simple consistency, in operations strategy, 20
- Skinner, Wickham, 6, 7, 11–15, 21

software, 15, 16–17, 19 <i>t</i> , 20, 22, 24–28, 25 <i>t</i> , 28 <i>t</i>	Target, 4
software companies, 23, 24	Tommy Hilfiger, 4
software development, 15, 25 <i>t</i>	unit costs, 12
software vendors, 6	virtual goods, 4
specialization, 13–14, 22, 23 <i>f</i>	Walmart, 4
specifications, 12	Walt Disney Company, 9–10
standardization, 11	warehouses, 16, 19 <i>t</i>
strategic alignment, 20, 24, 29	Wernerfelt, Birger, 15
strategy, 3, 4 <i>f</i> , 7, 8, 9 <i>f</i> , 11–12	Wheelwright, Steven, 11, 15
strategy hierarchy, 9, 9 <i>f</i>	workforce (decision area), 16, 22, 23 <i>f</i>
structural decisions, 21	working capital, 12
suppliers, 5, 13, 15, 17, 19 <i>t</i> , 22	
supply chain, 3, 4, 19 <i>t</i>	
supply chain management, 3, 17, 22, 26	